

TRIPOLI, Lebanon

WMO: 401030

Lat: 34.450N

Lon: 35.800E

Elev: 5

StdP: 101.26

Time Zone: 2.00 (LBN)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.6	6.8	-1.2	3.4	9.3	0.9	4.0	10.0	9.9	14.9	8.6	13.6	1.3	160	0.327

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.2	32.1	25.0	31.5	24.9	30.9	24.8	27.6	30.2	26.7	29.8	25.9	29.7	3.3	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.8	22.5	29.7	25.8	21.1	29.0	24.8	19.8	28.6	87.9	30.2	83.8	29.8	80.2	29.8	29.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.9	6.5	5.5	DB	3.1	34.5	2.1	1.2	1.6	35.4	0.4	36.1	-0.8	36.8	-2.3	37.7	(4)
			WB	1.5	27.4	2.3	1.3	-0.2	28.3	-1.6	29.1	-2.9	29.8	-4.6	30.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.1	12.6	13.4	15.3	17.9	21.3	24.5	26.8	27.6	26.3	23.0	18.4	14.1
	DBStd	5.66	2.33	2.29	2.43	2.22	2.07	1.51	1.07	0.86	1.30	2.19	2.41	2.37
	HDD10.0	13	7	3	1	0	0	0	0	0	0	0	0	3
	HDD18.3	612	179	139	97	33	2	0	0	0	0	1	28	132
	CDD10.0	3710	86	98	167	237	349	436	521	546	488	402	251	130
	CDD18.3	1267	0	0	5	21	92	186	263	288	238	144	29	1
	CDH23.3	11932	0	1	27	96	498	1556	2966	3479	2351	876	82	1
	CDH26.7	3823	0	0	4	17	81	347	1084	1383	749	154	3	0
Wind	WSAvg	2.1	2.2	2.2	2.3	2.3	2.2	2.4	2.4	2.3	2.1	1.6	1.5	1.9
Precipitation	PrecAvg	782	169	130	103	47	18	4	1	1	11	63	112	143
	PrecMax	1115	314	366	348	138	41	24	9	5	53	184	279	277
	PrecMin	464	35	36	16	12	6	2	0	0	0	4	5	36
	PrecStd	171	72	73	68	33	11	4	2	1	15	54	74	73
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.7	22.1	26.5	28.7	31.2	31.9	33.0	33.2	32.5	31.4	27.1	22.2
		MCWB	14.5	15.8	16.9	18.3	19.9	23.9	25.8	25.7	24.3	21.9	19.0	15.5
	2%	DB	18.9	20.2	23.0	25.8	28.7	30.1	31.8	32.1	31.4	29.5	25.4	20.5
		MCWB	14.0	14.9	16.2	18.4	20.4	23.6	25.5	25.4	24.3	21.8	18.0	15.0
	5%	DB	17.8	18.9	21.0	23.8	27.1	29.3	31.2	31.5	30.6	28.3	24.0	19.4
		MCWB	13.4	14.2	16.2	17.9	20.8	23.4	25.2	25.3	24.0	21.6	17.7	14.3
	10%	DB	16.8	17.8	19.6	22.5	25.9	28.6	30.6	31.1	30.0	27.4	22.8	18.3
		MCWB	12.6	13.5	15.2	17.4	20.5	23.1	25.0	25.2	23.6	21.2	17.1	13.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.1	16.8	18.9	20.6	23.7	26.2	28.3	28.8	27.5	25.0	20.5	17.2
		MCDB	18.4	20.3	22.9	25.6	26.5	28.7	30.8	30.9	30.0	27.8	25.2	20.7
	2%	WB	14.7	15.6	17.6	19.3	22.4	25.2	27.4	27.8	25.9	23.5	19.4	15.8
		MCDB	17.9	19.4	21.2	23.5	26.3	28.7	30.2	30.3	29.7	27.7	23.9	19.1
	5%	WB	13.8	14.7	16.5	18.5	21.5	24.6	26.6	26.9	25.1	22.7	18.6	15.0
		MCDB	17.1	18.4	20.3	22.5	25.7	28.4	29.8	29.9	29.5	27.1	22.9	18.2
	10%	WB	13.0	13.8	15.7	17.8	20.7	23.9	25.8	26.0	24.4	21.8	17.6	14.2
		MCDB	16.3	17.4	19.3	21.7	25.0	27.8	29.5	30.0	29.0	26.5	21.9	17.6
Mean Daily Temperature Range	5% DB	MDBR	7.2	7.6	8.1	8.3	8.3	7.6	7.5	7.2	7.6	8.5	8.6	7.7
		MCDBR	8.5	9.4	10.3	10.8	10.4	8.6	8.2	7.8	8.3	9.2	10.0	9.0
		MCWBR	5.4	6.0	6.4	6.3	6.4	5.5	5.0	4.6	4.9	5.4	5.7	5.4
	5% WB	MCDBR	8.0	9.0	9.3	9.6	9.0	8.3	7.7	7.3	7.8	8.0	9.0	8.1
		MCWBR	5.4	6.2	6.4	6.3	6.1	5.7	5.3	4.8	5.3	5.5	5.8	5.4
Clear-Sky Solar Irradiance	taub	0.379	0.411	0.454	0.504	0.496	0.437	0.452	0.461	0.441	0.452	0.399	0.384	
	taud	2.308	2.201	2.078	1.956	2.029	2.247	2.216	2.188	2.235	2.169	2.288	2.304	
	Ebn at Noon	810	822	816	792	803	850	834	821	819	768	779	778	
	Edh at Noon	103	127	155	183	173	139	142	144	131	129	104	98	
All-Sky Solar Radiation	RadAvg	2.45	3.32	4.79	6.01	7.27	8.09	7.91	7.14	5.99	4.35	3.10	2.31	
	RadStd	0.20	0.33	0.28	0.39	0.37	0.17	0.12	0.14	0.19	0.22	0.22	0.22	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (13 neighbors)	+0.30	N/A	-1.33	N/A	N/A	N/A	N/A	N/A	+173	+110

Nomenclature: See separate page